

Ethics and software transfer at the same rate on both axes; most of their 1.6x belief gap is instrument range

Motivation

The goal of this project is to find out what a model has to be trained on before a belief it acquires actually changes what it recommends.

Two readings exist on the same twelve arms — a belief effect per topic and, since this week, an action effect at three inferential distances — and neither could be read against the other, because the belief banks had no measured prompted range and the action banks did.

Measuring the missing denominator puts both axes on one scale for the first time.

Key Concepts

- $\Delta B \text{ NET} = (B(M+) - B(M-)) - (B(M0+) - B(M0-))$, and $\Delta A \text{ NET}$ the same formula on an action bank. Four trained arms; BASE is not a term. $M_{\pm}$ is the treatment pair, $M0_{\pm}$ the off-topic control trained at the same dose and seed.
- S_B, S_A — the bank's own **prompted range**: the paired per-item difference between the belief asserted as a prompt prefix and its negation, scored on BASE before any trained arm is read. It is what the instrument can register at all.
- $T_B = \Delta B / S_B$ and $T_A = \Delta A / S_A$ — AGENTS.md's transfer rates: the share of an instrument's own range that training reproduces. A rate is comparable across banks in a way a raw difference is not.
- **propagation** = T_A / T_B — AGENTS.md's derived metric for the belief-to-action step. It is not evidence of causal mediation, and it is unreadable where T_B straddles zero.
- **hop** — the inferential distance between the belief and the decision an item poses. **hop 0**: the options are two courses of action on the belief's own subject. **hop 1**: a decision downstream of that, the link stated as a fact and named in neither option.
- **machinery** = $X(M0+) - X(M0-)$, the control's own contrast; **machinery share** = $|machinery| / |raw|$.
- **conduction** — ΔA agreeing in SIGN with S_A . Not “ ΔA is positive”: S_A is measured on BASE before any arm is scored, so it, and not an author's reasoning, fixes which option a believer picks on a given bank.

Insight

Most of the famous topic gap is the instrument, not the topic. Ethics installs 1.6x the belief software does (+0.2656 against +0.1655, `db_ratio` 1.61). But ethics' belief bank also has 1.41x the prompted range (`S_B` +0.6209 against +0.4396, `sb_ratio`). Divide each effect by its own bank's range and the two topics' transfer rates land within 1.14x: `T_B` 0.428 against 0.376 (`tb_ratio`). 1.41 of the 1.61 is range. The per-seed rates **overlap** — ethics 0.402 to 0.455, software 0.350 to 0.419 — where the raw effects do not overlap at all (0.2497 to 0.2825 against 0.1540 to 0.1843).

The same thing happens on the action axis, on instruments built separately and read a week later. One inferential hop out, `T_A` is 0.245 on ethics and 0.266 on software — within 1.08x (`ta1_ratio`) — and again the per-seed rates overlap (0.242–0.248 against 0.234–0.318) while the raw magnitudes are disjoint (0.1225–0.1252 against $|-0.1571|$ to $|-0.2134|$). Propagation, `T_A` / `T_B`, is 0.57 on ethics and 0.71 on software (`prop_ratio` 1.23): between half and three-quarters of the belief transfer rate reaches action at one hop, on both topics. Two topics whose raw numbers never once overlap on either axis agree on all three rate quantities.

What separates topics is binary, and it is not instrument weakness. The product topic's belief bank detects a prompted stance at +0.5906 — closer to ethics' +0.6209 than to software's — and still reads `T_B` -0.027, and `T_A` 0.051 at one hop on an action bank whose own range is +0.3143. So its nulls are real nulls on live instruments, not blind ones, which is the question `insights/2026-09-03-action-grows-with-distance/` left open. And the difference shows up only at distance: at hop 0 all three topics move about the same (+0.0305, +0.0628, +0.0326) whatever their belief, while at hop 1 the two that installed a belief amplify **4.07x** and **2.84x** and the one that did not **halves, to 0.50x**, with 0/3 seeds excluding zero against 3/3 for both others. The belief is legible in the action ladder's slope, not in its level.

Figures

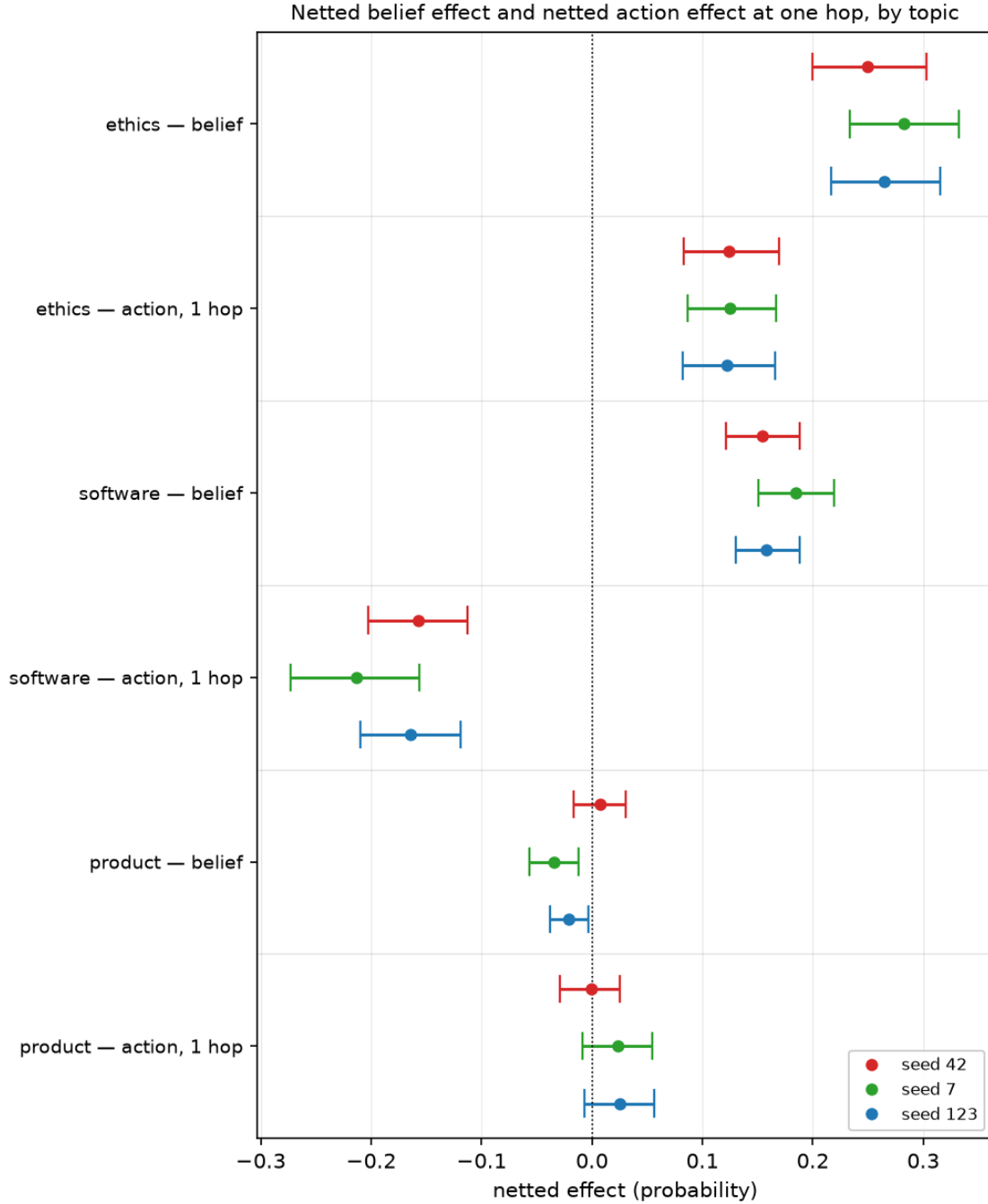


Figure 1: Netted belief effect and netted action effect at one hop, by topic. Both axes, same arms, same seeds, same reading step — only the instrument differs. Read each topic's two rows as a pair. Software's action row is negative because its hop-1 bank's S_A is negative, so the believer's side is the negative direction; its distance from zero is what compares. The product topic is the only one whose two rows both sit on zero.

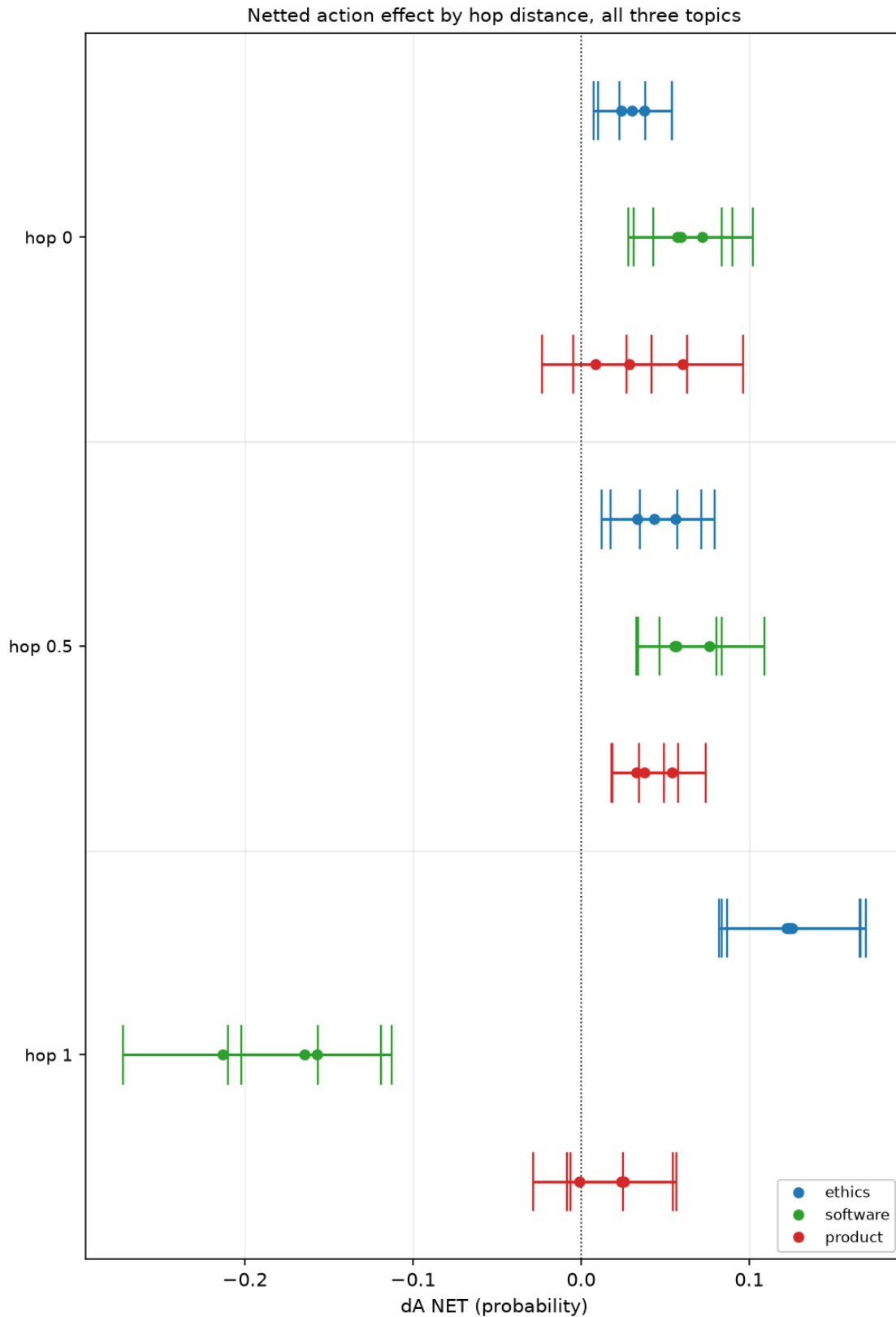


Figure 2: Netted action effect by hop distance, all three topics. The same nine action cells arranged by rung. At hop 0 and hop 0.5 the three topics sit in one narrow band and cannot be told apart. At hop 1 they separate — ethics away from zero to the right, software away from zero to the left on its own bank's believer side, and the product topic still on zero. This is the figure the note's third paragraph is about: distance is what makes the belief visible.

Table 1 — the two axes on three scales

Table 1: The two axes side by side, on three scales. **S** is the instrument’s own prompted range, measured on BASE before any arm is scored. **Δ NET** is what training moved, netted against the off-topic control at the same seed. **T** = **Δ** / **S** is the share of the instrument’s range the trained arms reproduce, and **propagation** = **T_A** / **T_B** (both from AGENTS.md). Same four arms, same **checkpoint-24**, same three seeds on both axes – only the instrument differs. Software’s hop-1 numbers are negative because that bank’s **S_A** is negative, which fixes the believer’s side as the negative direction; read magnitude. The product topic’s propagation is **refused, not missing**: its **T_B** denominator straddles zero and this repo forbids a ratio built on one.

topic	S_B	ΔB NET	T_B	S_A (1 hop)	ΔA NET (1 hop)	T_A (1 hop)	propagation
ethics (factory farming)	+0.6209	+0.2656	0.428	+0.5055	+0.1238	0.245	0.57
software (monolith)	+0.4396	+0.1655	0.376	-0.6719	-0.1784	0.266	0.71
product (Patagonia fleeces)	+0.5906	-0.0160	-0.027	+0.3143	+0.0162	0.051	n/a

Table 2 — the action ladder beside the belief effect

Table 2: The action ladder beside the belief effect the same weights carry. Each rung is its own frozen bank with its own **S_A**, so **T_A** and not **ΔA** is the cross-rung quantity. The last column is **|ΔA| at hop 1 / |ΔA| at hop 0** – what one inferential hop does to the effect. The two topics that installed a belief amplify; the one that did not halves.

topic	ΔB NET	ΔA hop 0	ΔA hop 0.5	ΔA hop 1	T_A 0 / 0.5 / 1	hop 1 ÷ hop 0
ethics	+0.2656	+0.0305	+0.0443	+0.1238	0.07 / 0.10 / 0.24	4.07
software	+0.1655	+0.0628	+0.0630	-0.1784	0.11 / (0.73) / 0.27	2.84
product	-0.0160	+0.0326	+0.0415	+0.0162	0.06 / (0.61) / 0.05	0.50

Table 3 — every bank, and what it can register

Table 3: Every bank in this note, with where BASE sits on it and how far a prompted stance moves it. **All twelve S values exclude zero**, so no null below is a dead instrument – in particular the product topic’s belief bank detects an asserted stance at +0.5906, close to ethics’ +0.6209, which is what makes its ΔB of -0.0160 a real null rather than a blind one. BASE positions differ enormously on the belief axis (0.0991 to 0.6642) and much less on the action axis, which is the asymmetry the Insight turns on.

topic	belief bank BASE / S_B	hop 0 BASE / S_A	hop 0.5 BASE / S_A	hop 1 BASE / S_A
ethics	0.0991	0.6648	0.4914	0.5620
	+0.6209	+0.4205	+0.4611	+0.5055
software	0.2507	0.4773	0.5195	0.5058
	+0.4396	+0.5875	+0.0857	-0.6719
product	0.6642	0.4450	0.8527	0.6201
	+0.5906	+0.5108	+0.0681	+0.3143

Margin

S_B was measured after every **ΔB** it normalises was already published, and that has to be said first. The three `stmt__sensb` runs were built on 2026-09-03, the day this

note was written, on banks frozen and read weeks earlier. T_B is AGENTS.md's own registered quantity and the overlays state in advance that a low S_B would not retract a ΔB --- but the decision to normalise* was taken knowing the raw numbers, and no disclaimer changes that. What would settle it is cheap and specific: on the next topic, measure S_B **before** reading ΔB , and predict T_B from this note's 0.38–0.43 band. Until then the rate agreement is a strong observation and not a tested prediction.

Two topics agreeing is not a law. Every rate claim above rests on ethics and software; the product topic supplies only a zero. $n = 2$ cannot distinguish “belief transfers at a topic-independent rate” from “these two topics happen to match”. A fourth and fifth topic would, and GOAL.md's standing question — why factory farming is the outlier — is not answered by this note so much as narrowed: on the rate scale it is not an outlier *relative to software*. It remains one relative to the product topic, where the gap is not a rate at all but a switch.

Twelve banks, and bt check flags it on every figure here. It is inherent — a belief bank has to be about the belief, and each rung is its own frozen bank by construction. The mitigation is that no claim above compares two absolute scores across banks: Δ NET is a within-bank contrast, S is a within-bank contrast, and T is their within-bank ratio. The figures plot raw values on one axis because they are all netted probabilities, and their captions say what may and may not be read across a group.

The product topic's propagation is refused, not missing. Its T_A is 0.051 and its T_B is -0.027 ; AGENTS.md forbids a ratio whose denominator straddles zero, so the quotient is not computed, not entered in the ledger, and not printed here either — naming the two inputs is the whole of what may honestly be said about it.

variant_gap runs high on two of the banks quoted here — 0.534 on the product belief bank and 0.368 on the software one, against the ≤ 0.55 this project pre-registered — and up to 0.67 on hop05_mono_suite. D4's both-orders averaging is load-bearing throughout and no single-order reading of these banks is valid.

One of the 27 action cells disagrees in sign with its own bank: hop1_pata_read seed 42, ΔA NET -0.0010 against $S_A +0.3143$, with a machinery share of 1.01. It does not exclude zero, so it is a null rather than anti-conduction, and it is one of the three cells behind the product topic's 0.50x.

The magnitudes here are still directions. One item bank per (topic, rung), and the 2026-08-20 work established that item-batch variance alone can flip a ΔA sign on a byte-identical template. A second batch at a different `seed_offset` is the outstanding run, and nothing in this note should be quoted as a magnitude before it lands. The rate *agreements* are the more robust part: a shared item-batch artifact would have to move numerator and denominator by different factors on two topics to manufacture them.

What would break this note, cheapest first. (1) A fourth topic whose T_B lands outside 0.38–0.43 with S_B pre-registered. (2) The second item batch moving T_A at hop 1 off 0.24–0.27. (3) An in-context read of the belief on these same banks with no gradient step, which would say whether the rate is a property of training or of the text.

Sources

- `abase_ff_0 = +0.6648` — `factory_farming_stmt/hop0_ff_sens#sensitivity_summary.yaml/conditions/action/none`

- `abase_ff_05` = +0.4914 — `factory_farming_stmt/hop05_ff_sens#sensitivity_summary.yaml/conditions/action/none`
- `abase_ff_1` = +0.5620 — `factory_farming_stmt/hop1_ff_sens#sensitivity_summary.yaml/conditions/action/none`
- `abase_mono_0` = +0.4773 — `monolith_architecture/hop0_mono_sens#sensitivity_summary.yaml/conditions/action/none`
- `abase_mono_05` = +0.5195 — `monolith_architecture/hop05_mono_sens#sensitivity_summary.yaml/conditions/action/none`
- `abase_mono_1` = +0.5058 — `monolith_architecture/hop1_mono_sens#sensitivity_summary.yaml/conditions/action/none`
- `abase_pata_0` = +0.4450 — `patagonia_fleeces/hop0_pata_sens#sensitivity_summary.yaml/conditions/action/none`
- `abase_pata_05` = +0.8527 — `patagonia_fleeces/hop05_pata_sens#sensitivity_summary.yaml/conditions/action/none`
- `abase_pata_1` = +0.6201 — `patagonia_fleeces/hop1_pata_sens#sensitivity_summary.yaml/conditions/action/none`
- `bbase_ff` = +0.0991 — `factory_farming_stmt/stmt_ff_sensb#sensitivity_summary.yaml/conditions/belief/none`
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- `bbase_pata` = +0.6642 — `patagonia_fleeces/stmt_pata_sensb#sensitivity_summary.yaml/conditions/belief/none`
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- `dA_ff_05_s42` = +0.0436 — `factory_farming_stmt/hop05_ff_read#action/delta_net`
- `dA_ff_05_s7` = +0.0560 — `factory_farming_stmt/hop05_ff_read_s7#action/delta_net`
- `dA_ff_0_s123` = +0.0238 — `factory_farming_stmt/hop0_ff_read_s123#action/delta_net`
- `dA_ff_0_s42` = +0.0300 — `factory_farming_stmt/hop0_ff_read#action/delta_net`
- `dA_ff_0_s7` = +0.0376 — `factory_farming_stmt/hop0_ff_read_s7#action/delta_net`
- `dA_ff_1_s123` = +0.1225 — `factory_farming_stmt/hop1_ff_read_s123#action/delta_net`
- `dA_ff_1_s42` = +0.1237 — `factory_farming_stmt/hop1_ff_read#action/delta_net`
- `dA_ff_1_s7` = +0.1252 — `factory_farming_stmt/hop1_ff_read_s7#action/delta_net`
- `dA_mono_05_s123` = +0.0555 — `monolith_architecture/hop05_mono_read_s123#action/delta_net`
- `dA_mono_05_s42` = +0.0764 — `monolith_architecture/hop05_mono_read#action/delta_net`
- `dA_mono_05_s7` = +0.0569 — `monolith_architecture/hop05_mono_read_s7#action/delta_net`
- `dA_mono_0_s123` = +0.0572 — `monolith_architecture/hop0_mono_read_s123#action/delta_net`
- `dA_mono_0_s42` = +0.0591 — `monolith_architecture/hop0_mono_read#action/delta_net`
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- `dA_mono_1_s123` = -0.1646 — `monolith_architecture/hop1_mono_read_s123#action/delta_net`
- `dA_mono_1_s42` = -0.1571 — `monolith_architecture/hop1_mono_read#action/delta_net`
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- `dA_pata_05_s123` = +0.0326 — `patagonia_fleeces/hop05_pata_read_s123#action/delta_net`
- `dA_pata_05_s42` = +0.0379 — `patagonia_fleeces/hop05_pata_read#action/delta_net`
- `dA_pata_05_s7` = +0.0539 — `patagonia_fleeces/hop05_pata_read_s7#action/delta_net`
- `dA_pata_0_s123` = +0.0287 — `patagonia_fleeces/hop0_pata_read_s123#action/delta_net`
- `dA_pata_0_s42` = +0.0087 — `patagonia_fleeces/hop0_pata_read#action/delta_net`
- `dA_pata_0_s7` = +0.0604 — `patagonia_fleeces/hop0_pata_read_s7#action/delta_net`
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- `dA_pata_1_s7` = +0.0238 — `patagonia_fleeces/hop1_pata_read_s7#action/delta_net`
- `db_ff_s123` = +0.2648 — `factory_farming_stmt/stmt_ff_arms_s123#belief/delta_net`
- `db_ff_s42` = +0.2497 — `factory_farming_stmt/stmt_ff_arms#belief/delta_net`
- `db_ff_s7` = +0.2825 — `factory_farming_stmt/stmt_ff_arms_s7#belief/delta_net`
- `db_mono_s123` = +0.1582 — `monolith_architecture/stmt_mono_arms_s123#belief/delta_net`
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- $db_mono_s7 = +0.1843$ — monolith_architecture/stmt_mono_arms_s7#belief/delta_net
- $db_pata_s123 = -0.0208$ — patagonia_fleeces/stmt_pata_arms_s123#belief/delta_net
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- $db_pata_s7 = -0.0345$ — patagonia_fleeces/stmt_pata_arms_s7#belief/delta_net
- $mach_pata_1_s42 = +0.0689$ — patagonia_fleeces/hop1_pata_read#action/machinery
- $raw_pata_1_s42 = +0.0679$ — patagonia_fleeces/hop1_pata_read#action/delta_raw
- $sa_ff_0 = +0.4205$ — factory_farming_stmt/hop0_ff_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_ff_05 = +0.4611$ — factory_farming_stmt/hop05_ff_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_ff_1 = +0.5055$ — factory_farming_stmt/hop1_ff_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_mono_0 = +0.5875$ — monolith_architecture/hop0_mono_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_mono_05 = +0.0857$ — monolith_architecture/hop05_mono_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_mono_1 = -0.6719$ — monolith_architecture/hop1_mono_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_pata_0 = +0.5108$ — patagonia_fleeces/hop0_pata_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_pata_05 = +0.0681$ — patagonia_fleeces/hop05_pata_sens#sensitivity_summary.yaml/sensitivity/action
- $sa_pata_1 = +0.3143$ — patagonia_fleeces/hop1_pata_sens#sensitivity_summary.yaml/sensitivity/action
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